

Skanda Bhairava

CV

About me

Computer Science student interested in distributed systems, and high performance computing. I enjoy learning by building tools and working on projects.

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 Bangalore, India

Languages

Python • Go • C++ • CUDA
• Rust • JS/TS

Tools

PyTorch • Ray • Docker • Svelte
• Gin • FastAPI • PyGame

Interests

- Distributed Systems
- High Performance Computing (HPC)
- Simulations
- Graphics
- Research

EXPERIENCE

June 2025 **Kaynes Technology**

SOFTWARE ENGINEER INTERN • Mysuru, India 

Designed and built internal tools for RFID (Radio Frequency Identification) reader configuration, DMI (Driver Machine Interface) testing and UI debugging. These tools were built for The Kavach project implemented in the Indian Railways.

EDUCATION

2023–Current **Bachelors •**

INFORMATION TECHNOLOGY (HONOURS)

Manipal Institute of Technology Bangalore 

CGPA: 9.26/10

HONOURS AND AWARDS

2026 **Best Research Poster Award, Turinger Symposium**

MANIPAL INSTITUTE OF TECHNOLOGY • Bangalore 

Received the Best Poster Award for research, at the Turinger Symposium, Manipal Institute of Technology Bangalore, February 2026.

2026 **Hackathon Finalist**

HCL GUVI • AI Impact Summit 2026, Delhi 

Our team was placed in the top 2% in the HCL GUVI hackathon for developing a HoneyPot to extract details from a Scammer in a textual environment.

2024, 2025 **Manipal Achievers Scholarship (Dean's List)**

MANIPAL ACADEMY OF HIGHER EDUCATION • Bangalore 

Received the Manipal Achievers Scholarship in 2024 and 2025 for maintaining a CGPA of 9.26 out of 10.

PUBLICATIONS

[First Author] Skanda M Bhairava, Gururaj H L. "ADaRES: Adaptive Data Reallocation and Epoch Scheduling in Federated Learning Under Network and Resource Constraints". COMSNETS 2026 doi.org/10.1109/COMSNETS67989.2026.11418093

ONGOING RESEARCH WORK

[First Author] . "Hierarchical Federated Learning for Edge Caching". Manuscript under review at a leading IEEE journal.

[First Author] . "Byzantine-Resistant Scalable Clustered Federated Learning".

PROJECTS (TOP 5)

HierarchicalFLSim

Python • Ray • Systems

Built a configurable Hierarchical Federated Learning simulator used to experiment with flexible topology, Non-IID distributions, client dropout and custom aggregation algorithms. The simulator also helps validate Cost model for hierarchical training.

DragonCommissions

Python • Automation, AWS • Management

Co-founded a plugin development hub that built 100+ custom plugins for gaming creators. Coordinated a team of around 20 developers, and handled client requirements for around 10+ creators. I also built internal automation tools and pipelines that made development workflows efficient.

Raytracer

C++ • Multithreading • Graphics

Implemented a 3D Ray Tracing Engine from scratch in C++ using hand-derived math, multithreading and a custom Pixel Representation Format.

Leaf Disease Detection

Python • Pytorch • ML

Developed a lightweight CNN that performed better than baseline architectures (AlexNet, DeepNet) on a leaf disease dataset in important metrics.

GravBall

Python • PyGame • Simulation

Designed and Optimized a Physics based Python game to run at 120 fps by redesigning collision and rendering loops.